

Unsteady Hydrodynamic Loading of Sluice Gates with Bottom Sills

بارگذاری هیدرودینامیکی ناپایدار دریچه‌های کشویی دارای آستانه

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Status

Preliminary concept prepared for discussion, subject to supervisor confirmation. Two points remain open: (1) a short recent-literature confirmation review of the gap; (2) confirmation that the scope does not overlap with current students' work. This concept is one half of a proposed paired programme and is independent of Zina's concept.

Title options

Discussion	Unsteady Hydrodynamic Loading of Sluice Gates with Bottom Sills
Formal PhD	Hydrodynamic Load Regimes of Sluice Gates with Bottom Sills: Mechanisms, Scaling and the Discharge-Efficiency versus Dynamic-Load Trade-off
Publication	Regime-based scaling of fluctuating hydrodynamic loads on sill-equipped sluice gates

01 PROBLEM

Adding a bottom sill under a gate is a known way to increase discharge capacity. In the supervisor's group, sills on **labyrinth sliding gates** increased the discharge coefficient by about 9.9–20.8%, depending on sill width [1]. The group's related work [1–3] characterises **time-averaged** performance. A sill also changes **where the flow separates** (gate lip, sill crest, or both), and therefore the **fluctuating** loads that drive gate vibration and fatigue. These loads appear not to have been quantified for sill-equipped gates, so an efficiency gain could carry an unmeasured dynamic cost.

02 RESEARCH GAP

The general physics of gate flow is **not** the gap: vortices, shear-layer oscillation below submerged gates and the effect of sills on **mean** pressure have been studied, as a targeted check confirmed. The narrower gap, which appears insufficiently addressed, concerns:

- a **regime-based** description of which hydraulic conditions produce strong fluctuating loads on sill-equipped gates;
- a **dimensionless relationship** linking sill geometry, opening and submergence to load magnitude and frequency content;
- a **criterion** identifying operating regions that keep the discharge gain without raising dynamic excitation.
- Fallback if the plain-sluice version is already covered: the same question for the group's labyrinth sliding gates with sills [1], for which no dynamic-load data were found.

03 THE SCIENTIFIC CHAIN

PHYSICAL MECHANISM	Which separation and shear-layer process dominates in each regime?
LOAD CHARACTERISATION	RMS magnitude, frequency content, narrow-band versus broadband loading
DIMENSIONLESS RELATION	Which groups, and which length scale, collapse the data?
ENGINEERING CRITERION	Operating regions with high discharge efficiency and low dynamic load

Measurements and simulations are **means** to discover the mechanism and the scaling; this is not a measurement campaign. The deliverable is the relationship and criterion, tested on held-out configurations.

04 HYDRAULIC REGIMES TO BE ESTABLISHED

Candidate regime (working hypothesis)	Expected dominant mechanism	What must be discovered
R1 Free flow, no or low sill	Separation at the gate lip; shear layer of the contracted jet	Baseline fluctuating-load level and spectral shape
R2 Free flow, effective sill	A second separation point at the sill crest; possible reattachment on the sill depending on its length	The sill height and length at which control of separation shifts from lip to sill
R3 Submerged flow	Recirculating roller over the jet; shear-layer feedback that may produce narrow-band oscillation	Whether submergence turns broadband loading into narrow-band loading, and at what submergence
R4 Transition free ↔ submerged	Unstable switching between R1/R2 and R3	Regime boundaries and whether the transition produces load peaks

05 DIMENSIONLESS FRAMEWORK (CANDIDATES, NOT FINAL)

Group	Role
$a/H1$ (relative opening)	Basic control variable
s/a , Ls/a , xs/a	Relative sill height, length and position: geometry controlling separation
Submergence ratio (tailwater / opening)	Free versus submerged regime
Fr (at the contracted section)	Inertial scaling of the jet
$St = f \cdot l/V$	Frequency scaling. The length scale l is to be determined: contracted jet thickness, lip–sill gap, or sill length
C_p , C_F (rms pressure and force coefficients)	Load magnitude normalised by jet dynamic pressure
Spectral concentration index	Share of load energy in a narrow band (more critical for resonance)
C_d ; Re	Hydraulic efficiency, paired with C_F for the trade-off; Re monitored to bound scale effects

06 WORKING HYPOTHESES (TO BE TESTED)

- Mechanism.** Above a threshold relative sill height, control of separation moves from the gate lip to the sill crest. That shift changes both the level and the spectral shape of the fluctuating load.
- Scaling.** Within each regime, the dominant load frequency collapses on a Strouhal number with a regime-specific length scale. Identifying that length scale is a primary outcome.
- Trade-off.** There is a range of sill geometry and operating conditions in which most of the discharge gain is kept while C_F and the spectral concentration stay at or below the no-sill reference.

07 PREDICTIVE MODELLING DIRECTION

No final equation is proposed in advance. The thesis must discover:

- the regime boundaries in the (s/a , submergence, $a/H1$) space;
- the length scale that collapses the dominant frequencies in each regime;
- the functional dependence of C_F and of spectral concentration on the sill groups, within each regime;
- whether one efficiency–load criterion holds across configurations, tested on held-out geometries.

The proposed model form is **semi-empirical and regime-conditioned**, with the structure set by the mechanism and coefficients fitted to data. Black-box machine learning is not the contribution. CFD explains mechanisms; it does not replace experiments.

08 METHODOLOGY AND BOUNDED SCOPE

Work package	Content	Output
WP1 Framework	Dimensional analysis, regime hypotheses, reduced test matrix	Test plan
WP2 Experiments	One rigid gate with 6–8 flush pressure transducers on the lip, skin plate and sill. Configurations: no sill plus 3 sill heights × 2 lengths (7 geometries) × 3 openings × 2–3 flow states (free, submerged), about 40–60 runs, a subset repeated	Load data; Paper J1
WP3 CFD	LES or DES with VOF at about 6–8 selected cases. Mesh verified with the Richardson/GCI procedure used in [3]. Validated against measured spectra	Mechanism evidence; Paper J2
WP4 Model and criterion	Regime map, scaling relationship and efficiency–load criterion, checked on held-out configurations. Prototype-scale interpretation with a scale-effect discussion	Criterion; Paper J3

09 VALIDATION

- Measured pressure spectra are the primary reference; CFD is accepted only where it reproduces them. The no-sill baseline is checked against classical underflow-gate data (compiled in the confirmation review).
- Predictive check: the relationship and criterion are derived on part of the matrix and tested on held-out configurations.

10 EXPECTED CONTRIBUTION AND PAPERS

If the hypotheses hold: a regime-based, dimensionless description of fluctuating loads on sill-equipped gates and a criterion linking the group's discharge-efficiency results to dynamic safety (novelty subject to the confirmation review).

Paper	Indicative content	Candidate journals
J1	Experimental fluctuating pressures and forces; regime identification	J. Hydraulic Research; J. Hydraulic Eng.
J2	Mechanism of separation control and Strouhal scaling (experiment + LES)	J. Fluids Struct.; Physics of Fluids
J3	Efficiency–dynamic-load criterion with prototype-scale interpretation	Water Resources Management; J. Irrig. Drain. Eng.

11 INDEPENDENCE

Rigid gate and pressure sensors only; no modal identification; nothing is needed from Zina's thesis. Laboratory requirements are listed in Document 01, §6.

References (supervisor's publications)

- [1] Abbaszadeh H., Tarinejad R. (2025). Soft computing and predicting labyrinth gates discharge coefficient. *Water Resources*, 52(4), 701–714. <https://doi.org/10.1134/S0097807824604734>
- [2] Abbaszadeh H., Tarinejad R., Roushangar K., Daneshfaraz R. (2026). Investigation of the effects of internal structure and material properties on the hydraulic performance of rectangular gabion weirs: experimental and numerical study. *Water Resources Management*, 40, 376. <https://doi.org/10.1007/s11269-026-04667-3>
- [3] Vaseti M.M., Akhtari A.A., Tarinejad R. (2026). Optimized mesh zoning for dam bottom outlet CFD simulations using Richardson extrapolation and grid convergence analysis. *Engineering Reports*, 8(6). <https://doi.org/10.1002/eng2.70800>